

Institute for Energy Technology (IFE)
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August 12, 2026

Editorial Office
Engineering Applications of Artificial Intelligence

Dear Editor,

We submit **“Partial Differential Equation Agents (PDE-Agents): A Large Language Model-Orchestrated Multi-Agent Framework for Automated Finite Element Simulations with Knowledge Graph-Augmented Reasoning”** for consideration as an original research article in Engineering Applications of Artificial Intelligence.

The manuscript introduces a framework in which three specialist large language model agents autonomously set up, run, analyse, and iteratively improve finite element simulations from natural-language descriptions. A graph-based retrieval-augmented generation knowledge base (Neo4j, 768-dimensional vector embeddings) grounds agent reasoning in curated material properties, failure patterns, and prior run lineage. We propose a novel *KG Smart* integration strategy and show that *integration pattern*, rather than knowledge content alone, determines whether retrieval augmentation improves agent performance.

The evaluation includes verification and validation confirming $\mathcal{O}(h^2)$ convergence, a three-way ablation across 50 tasks, a novel-material experiment (material property fidelity = 1.00 vs. 0.34 for the baseline), and production metrics from 1,369 runs (97.8% success). All code and data are available at <https://github.com/MatPro-IFE/pde-agents>.

To our knowledge, no prior work combines multi-agent large language model orchestration with production finite element solvers and graph-based retrieval for engineering simulation automation. The manuscript is original, is not under consideration elsewhere, and all authors have approved its submission.

Sincerely,

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